



CONDITION BASED ASSESSMENT

FULL SCANNING REPORT

SBUM DIAGNOSTIC PAHANG

TNB BAHAGIAN PEMBAHAGIAN PAHANG

FUNCTIONAL LOCATION (ERMS)	:	CCHL/PCE/J00155
FUNCTIONAL LOCATION (SITE)	:	CCHL/PCE/J00155
SUBSTATION NAME (ERMS)	:	S/S LOJI AIR HABU
SUBSTATION NAME (SITE)	:	SS LOJI AIR HABU
COORDINATE	:	4.438017, 101.390712
SUBSTATION TYPE	:	SS
AREA	:	CAMERON HIGHLAND
STATE	:	PAHANG
PO	:	42289580
DATE	:	01 JUN 2026

SCANNED BY	
THERMOGRAPHY	: MUHAMMAD ADAM BIN SAFARUDDIN (2017MY34N004)
ULTRASOUND	: MUHAMMAD ADAM BIN SAFARUDDIN (UET2016316)

EQUIPMENT/TOOLS	
THERMOGRAPHY	: FLIR T640 (SN 55912229) (CALIBRATION DUE DATE: 26 DEC 2026)
ULTRASOUND	: UE 10000 MPH (SN 105249) (CALIBRATION DUE DATE: 26 DEC 2026)

PREPARED BY	
NAME OF CONTRACTOR	: EET ENGINEERING SERVICES SDN BHD
ADDRESS	: NO.38, JALAN SS19/1D,
	: 47500, SUBANG JAYA, SELANGOR.

SUBSTATION SIGNBOARD



EXECUTIVE SUMMARY

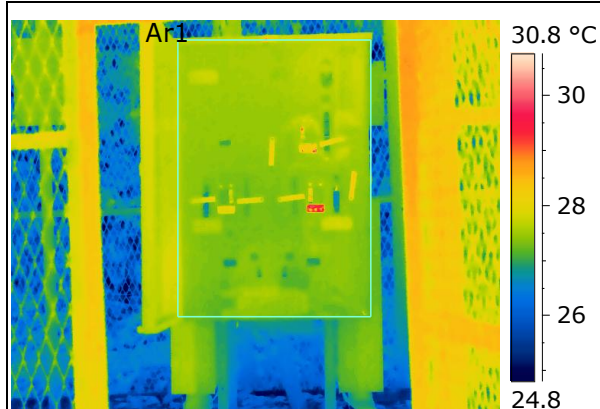
NO.	EQUIPMENT	AREA	IR (Abs T/ $\hat{a}^{\dagger}$ T)	US (dB)	DEFECT
1.	RMU SF6 TAMCO	OVERVIEW	-	-	-
		OVERVIEW TOP	-	-	-
		OVERVIEW BOTTOM	-	-	-
	PANEL CCH00347 KTRS TNB	CABLE COMPARTMENT	-	-	
		CABLE ENTRY	-	-	
	PANEL CCH00346 LOJI AIR	CABLE COMPARTMENT	-	-	
		CABLE ENTRY	-	-	
	PANEL CCH00345 SSU HABU	CABLE COMPARTMENT	-	-	
		CABLE ENTRY	-	-	

VISUAL DEFECT SUMMARY

NO.	EQUIPMENT	DEFECT DESCRIPTION	ADDITIONAL REMARKS
1	SUBSTATION	CPR POSTER OLD VERSION	SWG BAY

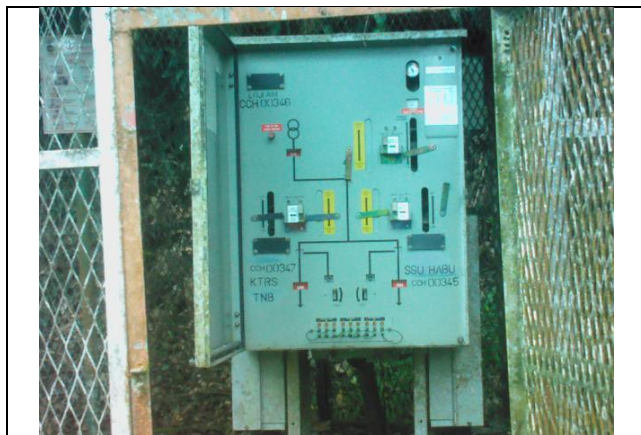
	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	OVERVIEW		Panel No.	-
Panel Name	-				

IR IMAGE



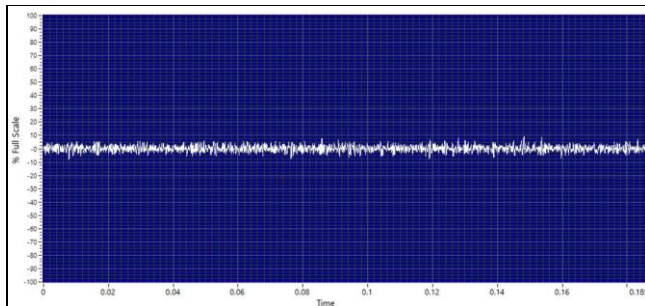
PARAMETER	VALUE
Spot Temp.1 (Sp1)	-
Spot Temp. 2 (Sp2)	-
Average Temp. (Ar1)	27.4 °C
Ambient Temp.	21.2 °C
Delta T (Δ T)	-
Load (A)	-
Defect	-

VISUAL IMAGE



PARAMETER	VALUE
Serial No.	B971569
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	-

ULTRASOUND IMAGE



Decibel (dB)	-
Sound characteristic	-
Severity	-
Sound clip	

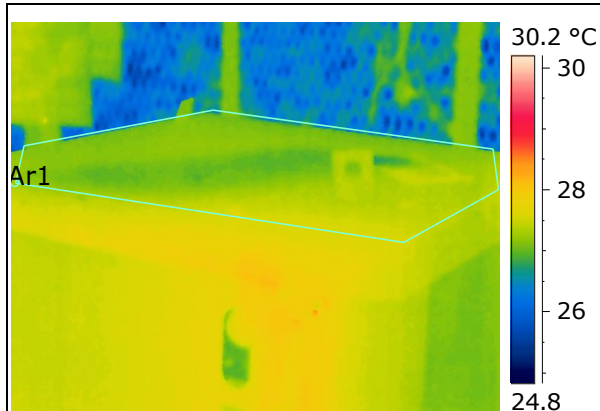
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	OVERVIEW TOP		Panel No.	-
Panel Name	-				

IR IMAGE



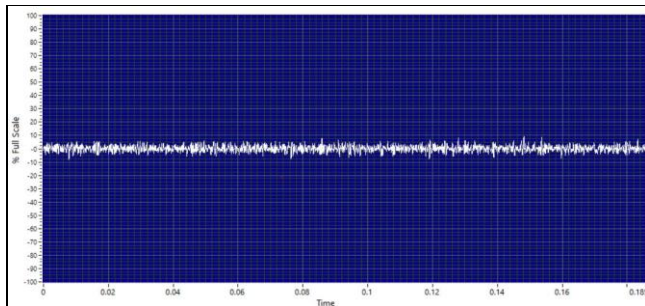
PARAMETER	VALUE
Spot Temp.1 (Sp1)	-
Spot Temp. 2 (Sp2)	-
Average Temp. (Ar1)	27.2 °C
Ambient Temp.	21.2 °C
Delta T (ΔT)	-
Load (A)	-
Defect	-

VISUAL IMAGE



PARAMETER	VALUE
Serial No.	B971569
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	-

ULTRASOUND IMAGE



Decibel (dB)	-
Sound characteristic	-
Severity	-
Sound clip	

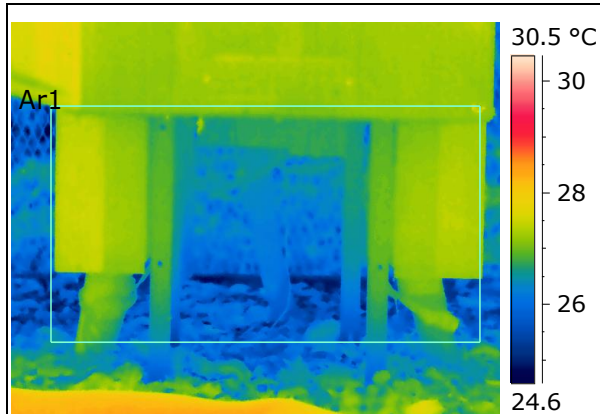
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	OVERVIEW BOTTOM		Panel No.	-
Panel Name	-				

IR IMAGE



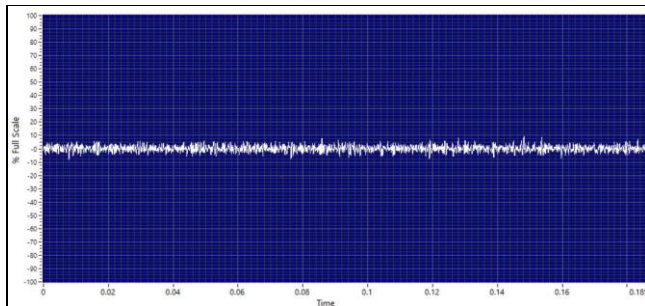
PARAMETER	VALUE
Spot Temp.1 (Sp1)	-
Spot Temp. 2 (Sp2)	-
Average Temp. (Ar1)	26.6 °C
Ambient Temp.	21.2 °C
Delta T (ΔT)	-
Load (A)	-
Defect	-

VISUAL IMAGE



PARAMETER	VALUE
Serial No.	B971569
Heater (A)	-
CB Position	-
Busbar Position	-
Cable Type	-

ULTRASOUND IMAGE



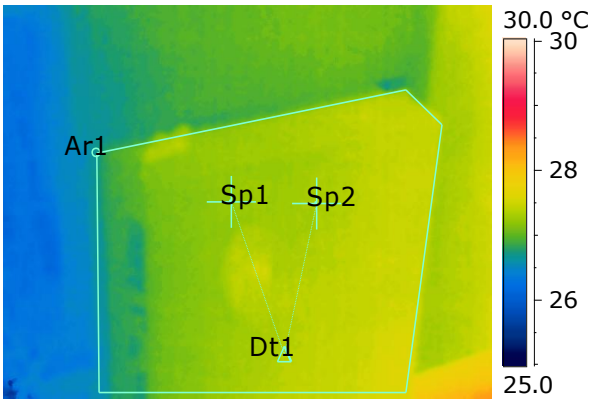
Decibel (dB)	-
Sound characteristic	-
Severity	-
Sound clip	

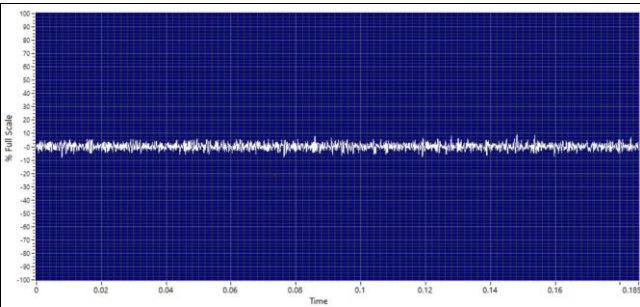
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	CABLE COMPARTMENT		Panel No.	CCH00347
Panel Name	KTRS TNB				

IR IMAGE		VISUAL IMAGE	
			
PARAMETER	VALUE	PARAMETER	VALUE
Spot Temp.1 (Sp1)	27.2 °C	Serial No.	B971569
Spot Temp. 2 (Sp2)	27.3 °C	Heater (A)	-
Average Temp. (Ar1)	27.2 °C	CB Position	CLOSE
Ambient Temp.	21.2 °C	Busbar Position	-
Delta T (Δ T)	0.1 °C	Cable Type	XLPE
Load (A)	-		
Defect			

ULTRASOUND IMAGE	
	
Decibel (dB)	0dB
Sound characteristic	Normal
Severity	
Sound clip	

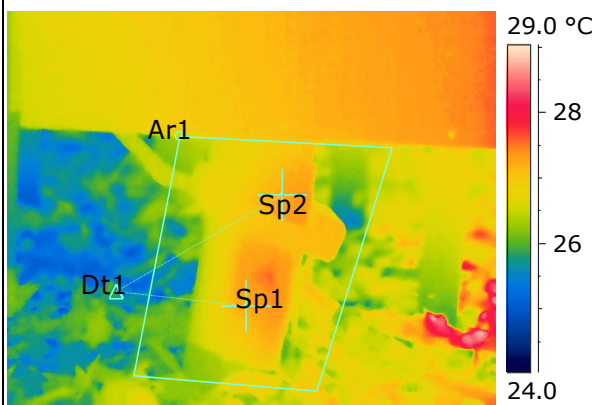
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	CABLE ENTRY		Panel No.	CCH00347
Panel Name	KTRS TNB				

IR IMAGE



PARAMETER

VALUE

Spot Temp.1 (Sp1)

27.2 °C

Spot Temp. 2 (Sp2)

27.2 °C

Average Temp. (Ar1)

26.6 °C

Ambient Temp.

21.2 °C

Delta T (Δ T)

0.0 °C

Load (A)

-

Defect

VISUAL IMAGE



PARAMETER

VALUE

Serial No.

B971569

Heater (A)

-

CB Position

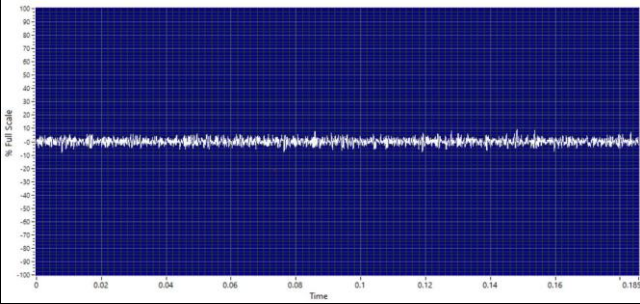
CLOSE

Busbar Position

-

Cable Type

XLPE

ULTRASOUND IMAGE	
	
Decibel (dB)	0dB
Sound characteristic	Normal
Severity	
Sound clip	

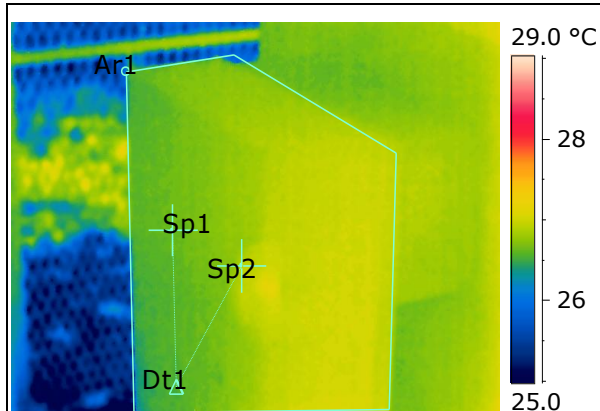
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	CABLE COMPARTMENT	Panel No.	CCH00346	
Panel Name	LOJI AIR				

IR IMAGE



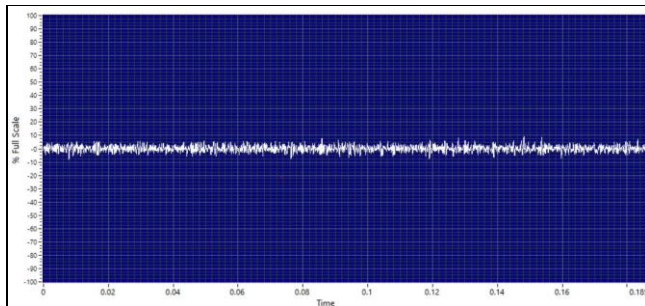
PARAMETER	VALUE
Spot Temp.1 (Sp1)	26.6 °C
Spot Temp. 2 (Sp2)	26.9 °C
Average Temp. (Ar1)	26.8 °C
Ambient Temp.	21.2 °C
Delta T (ΔT)	0.3 °C
Load (A)	-
Defect	

VISUAL IMAGE



PARAMETER	VALUE
Serial No.	B971569
Heater (A)	-
CB Position	CLOSE
Busbar Position	-
Cable Type	XLPE

ULTRASOUND IMAGE



Decibel (dB)	0dB
Sound characteristic	Normal
Severity	
Sound clip	

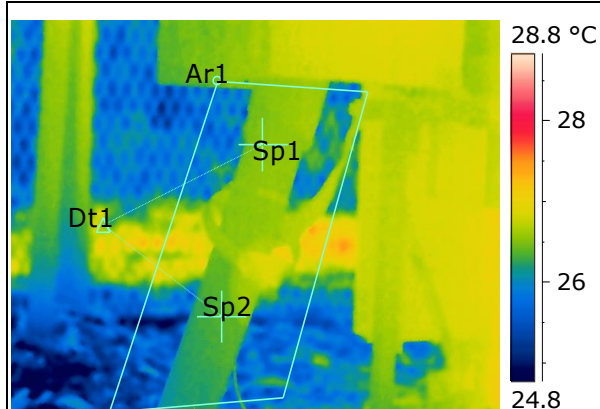
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	CABLE ENTRY		Panel No.	CCH00346
Panel Name	LOJI AIR				

IR IMAGE



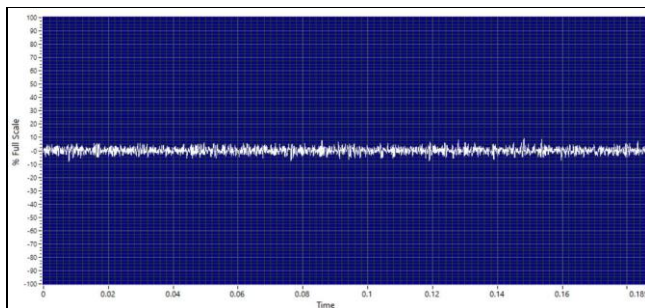
PARAMETER	VALUE
Spot Temp.1 (Sp1)	26.5 °C
Spot Temp. 2 (Sp2)	26.4 °C
Average Temp. (Ar1)	26.3 °C
Ambient Temp.	21.2 °C
Delta T (ΔT)	0.1 °C
Load (A)	-
Defect	

VISUAL IMAGE



PARAMETER	VALUE
Serial No.	B971569
Heater (A)	-
CB Position	CLOSE
Busbar Position	-
Cable Type	XLPE

ULTRASOUND IMAGE



Decibel (dB)	0dB
Sound characteristic	Normal
Severity	
Sound clip	

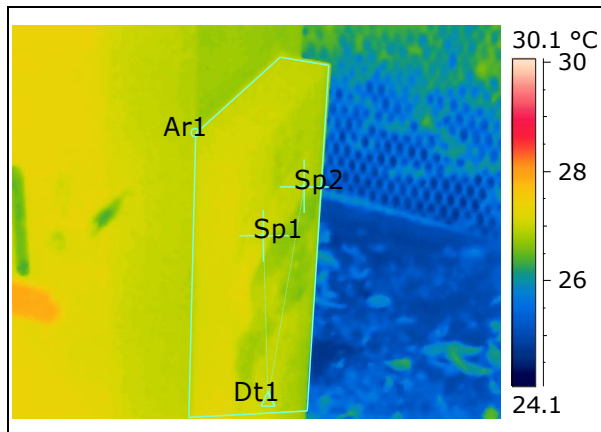
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	CABLE COMPARTMENT		Panel No.	CCH00345
Panel Name	SSU HABU				

IR IMAGE



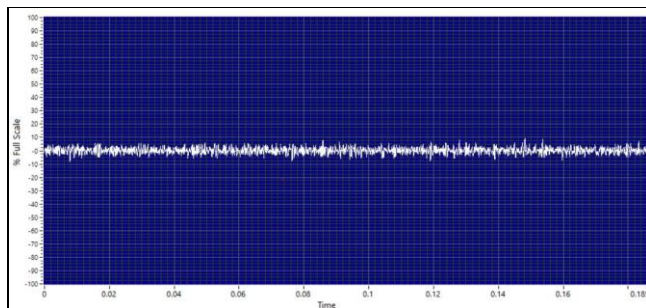
PARAMETER	VALUE
Spot Temp.1 (Sp1)	27.0 °C
Spot Temp. 2 (Sp2)	26.9 °C
Average Temp. (Ar1)	27.0 °C
Ambient Temp.	21.2 °C
Delta T (ΔT)	0.2 °C
Load (A)	-
Defect	

VISUAL IMAGE



PARAMETER	VALUE
Serial No.	B971569
Heater (A)	-
CB Position	CLOSE
Busbar Position	-
Cable Type	XLPE

ULTRASOUND IMAGE



Decibel (dB)	0dB
Sound characteristic	Normal
Severity	
Sound clip	

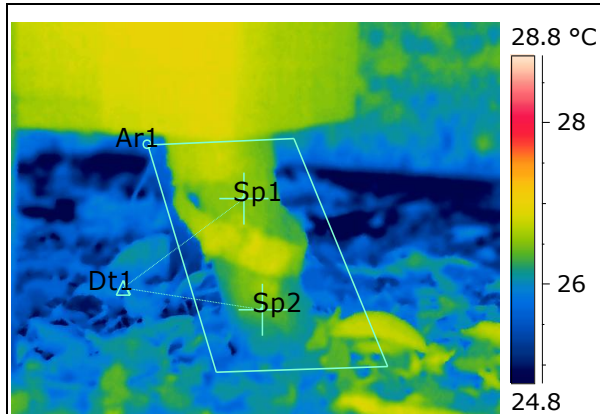
Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

	Substation	S/S LOJI AIR HABU			Date: 01/06/2026
	Equipment	RMU SF6	Manufacturer	TAMCO	Time: 4:21 PM
	Model	FALCON BETA	Rating	12kV	Humidity: 80%
	Area	CABLE ENTRY		Panel No.	CCH00345
Panel Name	SSU HABU				

IR IMAGE



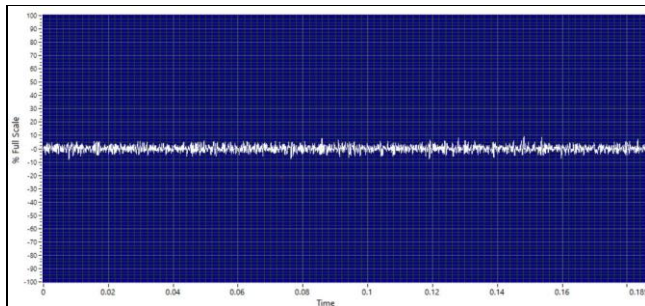
PARAMETER	VALUE
Spot Temp.1 (Sp1)	26.5 °C
Spot Temp. 2 (Sp2)	26.4 °C
Average Temp. (Ar1)	26.2 °C
Ambient Temp.	21.2 °C
Delta T (ΔT)	0.1 °C
Load (A)	-
Defect	

VISUAL IMAGE



PARAMETER	VALUE
Serial No.	B971569
Heater (A)	-
CB Position	CLOSE
Busbar Position	-
Cable Type	XLPE

ULTRASOUND IMAGE



Decibel (dB)	0dB
Sound characteristic	Normal
Severity	
Sound clip	

Analysis & Recommendations:

Analysis: No Anomaly.

Recommendation: -

SUBSTATION CONDITION

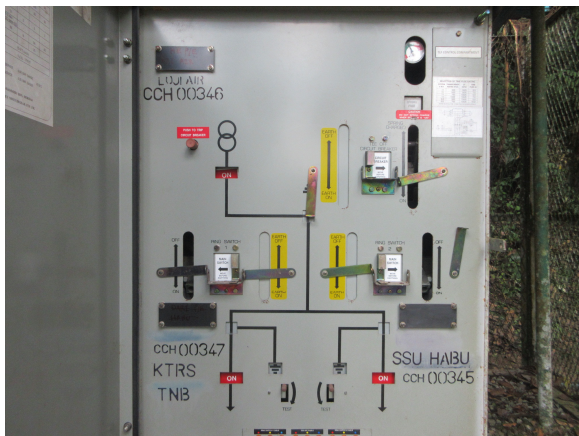
SUBSTATION OVERVIEW



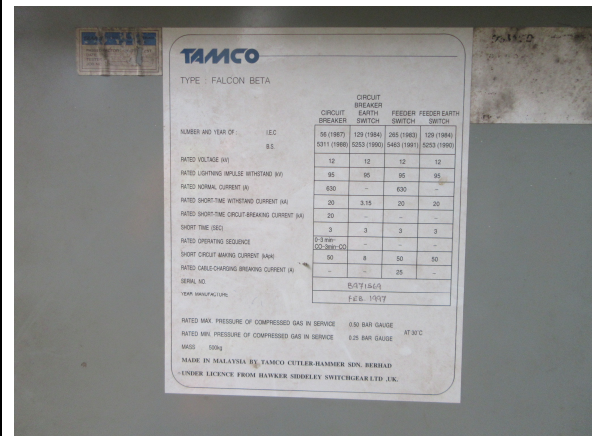
SIGNBOARD



SWITCHGEAR



SWITCHGEAR NAMEPLATE



EFI



SF6 INDICATOR



VISUAL DEFECT



CPR POSTER OLD VERSION – SWG BAY

NORMAL/DEFECT STICKER

NORMAL STICKER	
CONDITION ASSESSMENT TENAGA NASIONAL Better Together DIAGNOSTIC UNIT SUBSTATION 	
Cycle/Year :	1/2026
Testing Date :	01-06-2026
Defect :	—
Company/Team :	EET ENGINEERING
	TEV U/S IR

152
SS

STATE / AREA: PAHANG / CAMERON HIGHLAND										DISTRIBUTION SUBSTATION (PCS) CBA TEST SHEET										DATE: 01/06/2026										FUNCTIONAL LOCATION: CCHL/PCB/1700155																			
PE NAME: SIS LOJI AIR HABU										TIME IN / OUT: 4:21 pm - 4:34 pm										TEV BACK GROUND / HUMIDITY: 3 - 80%										BACKGROUND TEMP: 21.2 °C																			
V.I REF. NO:										SWITCHGEAR										TEMPERATURE										ULTRASOUND										TEV									
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLIV)	COMP	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE / CYCLE	TYPE CHAR.	REMARK																												
1	CCH 00347	KTR5 TNB	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	27.0	27.2	0.2	27.1	39.30		0	7	-	4	0	-																													
2	CCH 00346	LOJI AIR	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	26.5	26.9	0.4	26.8	31.32		0	8	-	4	0	-																													
3	CCH 00345	SSV HABU	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	26.4	26.7	0.3	26.5	33.34		0	9	-	5	0	-																													
4			TRIP / CLOSE		XLPE PILC N/A NACC			CABLE																																									
OVERVIEW										bottom										u6																													

LOCAL TRANSFORMER										THERMOGRAPHY										ULTRASOUND										THERMOGRAPHY										ULTRASOUND									
TX ID	SERIAL NO	CABLE TYPE/ SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	AT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/ SIZE	COMP.	Tmin(°C)	Tmax(°C)	AT	FILE	Avg (°C)	FILE	dB	TYPE CHAR.	REMARKS																									
TX1	N/A	XLPE/PILC		HT CABLE								TX3		XLPE/PILC	HT CABLE																																		
		XLPE/PILC		HT BUSHING										XLPE/PILC	HT BUSHING																																		
		XLPE/PILC		LV CABLE										XLPE/PILC	LV CABLE																																		
		XLPE/PILC		LV BUSHING										XLPE/PILC	LV BUSHING																																		
		XLPE/PILC		BODY										XLPE/PILC	BODY																																		
		XLPE/PILC		HT CABLE										XLPE/PILC	HT CABLE																																		
		XLPE/PILC		HT BUSHING										XLPE/PILC	HT BUSHING																																		
		XLPE/PILC		LV CABLE										XLPE/PILC	LV CABLE																																		
		XLPE/PILC		LV BUSHING										XLPE/PILC	LV BUSHING																																		
		XLPE/PILC		BODY										XLPE/PILC	BODY																																		

LV DISTRIBUTION BOARD / FEEDER PILLAR / OTHERS										THERMOGRAPHY										ULTRASOUND										THERMOGRAPHY										ULTRASOUND									
NO	LVDB / PP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	AT	FILE	Avg (°C)	REMARK	LVDB / PP 1	J-SLOTTED / DIN TYPE	Manufacturer:	Serial No.:	Rating:	LVDB / PP 2	J-SLOTTED / DIN TYPE	Manufacturer:	Serial No.:	Rating:																															
1	LVDB / PP 1	INC. 1							F1					F2																																			
2	LVDB / PP 2	INC. 2							F3					F4																																			
3	LVDB / PP 3	INC. 3							F5					F6																																			
4	LVDB / PP 4	INC. 4							F7					F8																																			
5	LVDB / PP 5	INC. 5							F9					F10																																			
6	LVDB / PP 6	INC. 6							F11					F12																																			
7	LVDB / PP 7	INC. 7							F13					F14																																			
8	LVDB / PP 8	INC. 8							F15					F16																																			
9	LVDB / PP 9	INC. 9							F17					F18																																			
10	LVDB / PP 10	INC. 10							F19					F20																																			
11	LVDB / PP 11	INC. 11							F21					F22																																			
12	LVDB / PP 12	INC. 12							F23					F24																																			
13	LVDB / PP 13	INC. 13							F25					F26																																			
14	LVDB / PP 14	INC. 14							F27					F28																																			
15	LVDB / PP 15	INC. 15							F29					F30																																			
16	LVDB / PP 16	INC. 16							F31					F32																																			
17	LVDB / PP 17	INC. 17							F33					F34																																			
18	LVDB / PP 18	INC. 18							F35					F36																																			
19	LVDB / PP 19	INC. 19							F37					F38																																			
20	LVDB / PP 20	INC. 20							F39					F40																																			
21	LVDB / PP 21	INC. 21							F41					F42																																			
22	LVDB / PP 22	INC. 22							F43					F44																																			
23	LVDB / PP 23	INC. 23							F45					F46																																			
24	LVDB / PP 24	INC. 24							F47					F48																																			
25	LVDB / PP 25	INC. 25							F49					F50																																			
26	LVDB / PP 26	INC. 26							F51					F52																																			
27	LVDB / PP 27	INC. 27							F53					F54																																			
28	LVDB / PP 28	INC. 28							F55					F56																																			
29	LVDB / PP 29	INC. 29							F57					F58																																			
30	LVDB / PP 30	INC. 30							F59					F60																																			
31	LVDB / PP 31	INC. 31							F61					F62																																			
32	LVDB / PP 32	INC. 32							F63					F64																																			
33	LVDB / PP 33	INC. 33							F65					F66																																			
34	LVDB / PP 34	INC. 34							F67					F68																																			
35	LVDB / PP 35	INC. 35							F69					F70																																			
36	LVDB / PP 36	INC. 36							F71					F72																																			
37	LVDB / PP 37	INC. 37							F73					F74																																			
38	LVDB / PP 38	INC. 38							F75					F76																																			
39	LVDB / PP 39	INC. 39							F77					F78																																			
40	LVDB / PP 40	INC. 40							F79					F80																																			
41	LVDB / PP 41	INC. 41							F81					F82																																			
42	LVDB / PP 42	INC. 42							F83					F84																																			
43	LVDB / PP 43	INC. 43							F85					F86																																			
44	LVDB / PP 44	INC. 44							F87					F88																																			
45	LVDB / PP 45	INC. 45							F89					F90																																			
46	LVDB / PP 46	INC. 46							F91					F92																																			
47	LVDB / PP 47	INC. 47							F93					F94																																			
48	LVDB / PP 48	INC. 48							F95					F96																																			
49	LVDB / PP 49	INC. 49							F97					F98																																			
50	LVDB / PP 50	INC. 50							F99					F100																																			

EET ENGINEERING SERVICES SDN BHD
 No. 38-2, 2nd Floor, Jalan SS 19/1D
 47500 Subang Jaya, Selangor.
 Tel: 603-5636 6205 Fax: 603-5636 7205

MOHD AZZAD BIN OTHMAN (10085230)
 JURUTEKNIK TINGKATAN BIASA
 (DIAGNOSTIC SUBSTATION PAHANG)
 SBU MAINTENANCE EAST
 DISTRIBUTION NETWORK DIVISION, TNB

152 SS

LPA COV. Sny Dug 1



VISUAL INSPECTION CHECKLIST FOR 11kV SUBSTATION

DATE: 1/6/2026
P/E Name (ERMS): S/S LOJI AIR HABU
P/E Name (SITE): SS LOJI AIR HABU
GPS Coordinate: 4.438017, 101.390712
PCE Type: INDOOR
PCE Area: NORMAL
Switchgear Type 1: RMU OIL
Manufacturer/Modal: TAMCO / FALCON BETA
Switchgear Manufactured Date: 1997
Switchgear Type 2: RMU OIL
Manufacturer/Modal: N/A
Switchgear Manufactured Date: N/A
No. of Transformer: N/A
Tx 1: Type
Tx 2: Type
Tx 3: Type
Tx 4: Type

F/L NO (ERMS): CCLAL/PCE/300155
F/L NO (SITE): CCLAL/PCE/300155

OUTDOOR
INDUSTRIAL
RMU SF6
COMPACT
COASTAL
MRMU
ATTACH BUILDING
PAT
VCB
OCB

Switchgear Rating: 12kV
RMU SF6 / OIL S/N: B971569

Switchgear Rating:
RMU SF6 / OIL S/N:
Construction Yr:
Manufacturer:
S/N:
Construction Yr:
Manufacturer:
S/N:
Construction Yr:
Manufacturer:
S/N:

KEJANGGALAN

1. Signboard:
2. Earth connection:
3. EFI Functioning:
4. Heater Functioning:
5. Labelling @ Switchgear:
6. Labelling @ LVDB:
7. Gas Pressure:
8. Tx Guard:
9. Tx Oil Level (conservator tank):
10. Tx Oil Leak:
11. Tx Bushing:
12. Top Demand Indicator (@LVDB):
13. Link/fuse Holder:
14. LV Termination:
15. Rubber Mat:
16. Trenching:
17. Battery Voltage:
18. Lighting:
19. Roof Condition:
20. Fire Fighting
21. Fence/door condition:
22. Cleanliness/Bushes:
23. Vandalisme
24. Other Findings

YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Valid	✓	Expired		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO		Remarks

TESTED BY (NAME & SIGNATURE):

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Tel: 603-5636 6205 Fax: 603-5636 7205

TNB SUPERVISOR/ ENGINEER (NAME & SIGNATURE):

MOHD AZZAD BIN OTHMAN (10085230)
JURUTEKNIK TINGKATAN BIASA
(DIAGNOSTIC SUBSTATION PAHANG)
SBU MAINTENANCE EAST
DISTRIBUTION NETWORK DIVISION, TNB